

MATH8009: Linear Algebra - Problem Sheet 9

Q1. Find the solution set S to the following 2×2 linear systems using the *Inverse Matrix Method*:

a)
$$\begin{aligned} 2x + 3y &= 5 \\ x + 4y &= 10 \end{aligned}$$

b)
$$\begin{aligned} 3x - 6y &= 8 \\ 2x + 5y &= 1 \end{aligned}$$

c)
$$\begin{aligned} x + 2y &= 7 \\ 2x + 5y &= -3 \end{aligned}$$

d)
$$\begin{aligned} 4x + 2y &= 0 \\ -x - 3y &= 0 \end{aligned}$$

Q2. Find the solution set S to the above 2×2 linear systems in Q4 using the *Gauss-Jordan Method*.

Q3. Determine the solution set S to the following 2×2 linear systems (using either method)

a)
$$\begin{aligned} x + 2y &= 3 \\ -x + 4y &= 8 \end{aligned}$$

b)
$$\begin{aligned} x + 2y &= 3 \\ -x - 4y &= 6 \end{aligned}$$

c)
$$\begin{aligned} x + 2y &= 3 \\ -x - 4y &= -6 \end{aligned}$$

Q4. Find the solution set to the following 3×3 systems of linear equations using the Gauss-Jordan Method:

a)
$$\begin{aligned} x + y + z &= 6 \\ 2x + 3y - 2z &= 2 \\ 3x - 2y + 4z &= 11 \end{aligned}$$

b)
$$\begin{aligned} 2x + 3y - z &= -5 \\ -6x + y + 3z &= 5 \\ 3x - 5y - 2z &= -1 \end{aligned}$$

c)
$$\begin{aligned} 3x + 5y - 2z &= 0 \\ 2x - 3y - 2z &= -5 \\ x + y + z &= -4 \end{aligned}$$

d)
$$\begin{aligned} -2x + 3y + z &= 6 \\ 3x - 5y + 2z &= 7 \\ x + y - z &= 2 \end{aligned}$$

Q5. The following augmented matrices have been row reduced using row operations to echelon form. Identify if any of the variables x_1, x_2, x_3 , etc are free and write down the solution set S for each system.

a)
$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 5 \end{array} \right)$$

b)
$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & -2 \\ 0 & 1 & 4 & 1 \end{array} \right)$$

c)
$$\left(\begin{array}{ccc|c} 1 & 6 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right)$$

d)
$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 0 & 6 \\ 0 & 1 & 4 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \end{array} \right)$$

e)
$$\left(\begin{array}{cc|c} 1 & 2 & 8 \\ 0 & 0 & 0 \end{array} \right)$$

f)
$$\left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 3 \end{array} \right)$$

g)
$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 4 & -1 \\ 0 & 1 & 0 & 2 & 6 \\ 0 & 0 & 1 & 3 & 8 \end{array} \right)$$

h)
$$\left(\begin{array}{ccccc|c} 1 & 3 & 0 & 0 & 4 & -2 \\ 0 & 0 & 1 & 0 & -2 & 1 \\ 0 & 0 & 0 & 1 & 5 & 3 \end{array} \right)$$